Manuel Rodríguez Villegas

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Profile

AI engineer with international research and industry experience and a strong mathematical background. My approach starts with first-principles reasoning to turn complex problems into simple, efficient solutions. I bring energy and enjoy fast-paced environments where I can take ownership and meaningful responsibility.

Education

Comillas Pontifical University, ETSI ICAI — Madrid, Spain
Master's Degree in Artificial Intelligence; GPA: 9.08/10; Class Representative — *2025 – 2026*
- Ongoing Master's thesis: Vision-only autonomous drones with differentiable physics.

Cornell University — Ithaca, NY, USA
Exchange Student - Electrical & Computer Engineering — *Jan – May 2025*

Comillas Pontifical University, ETSI ICAI — Madrid, Spain
BE in Mathematical Engineering and AI; GPA: 8.75/10 — *2021 – 2025*

Experience

AUDI AG — Ingolstadt, Germany
AI Engineering Intern — *July 2026 – Nov 2026*
- Developing 3D defect classification under limited data and strong confounding.
- Building AI agent skills for NLP-based requirements classification and conflict detection, and reliable, low-latency, and token-efficient access to a graph database with 100,000+ records.

Comillas Pontifical University — Madrid, Spain
Teaching Assistant — *Sep 2025 – Apr 2026*
- Contributed to deployment of a GPU-based AI cluster to support student workloads in deep learning courses.

Azzulei Technologies — Madrid, Spain
Computer Vision Intern — *Jun – Aug 2025*
- Built a real-time multi-object tracking pipeline (YOLOv8 + Optical Flow) for automated sports camera control, achieving robust performance at 30 FPS.

Imperial College London — Remote (London, UK)
Research Assistant — *Jun – Aug 2024*
- Reproduced nonlinear dynamical systems using Neural ODEs and PINNs (PyTorch), validating stability and trajectory accuracy against ground-truth simulations and different initial conditions.

Endesa — Madrid, Spain
Energy Data Analyst Intern — *Jun – Aug 2023*
- Forecasted daily Iberian ancillary service costs using MLP and regression models (MAPE $< 10\%$).

Selected Research

TopoSIGMA: Spectral Information-Geometric Mapping Architecture — 2026
- Designed an online, training-free ROS 2 mapping system separating spectral place discovery from covariance-aware loop closure, using DINOv2 descriptors and directional node distributions.
- Manuscript in preparation for submission to Robotics and Autonomous Systems.

Skills

Software: Python (advanced), PyTorch, JAX, ROS 2, Git, Docker, K8S, LangGraph, n8n
AI & ML: Deep Learning, Generative & Agentic AI (MCP), Geometric AI, Computer Vision, NLP, MLOps, DRL
Mathematics: Linear Algebra, Probabilistic AI, Optimization, Differential Geometry, Statistics

Languages

Spanish (Native) | English (Fluent) | German (Beginner)

Honors & Awards

Bending Spoons Hackathon Winner — Milan, Italy | 2026
Selected among Spain's top 20 tech students for First Ascent and winner of the event's hackathon.

Hackathon Winner — Madrid, Spain | 2025
Developed the winning project in the 10th Smart Industry Hackathon, featuring Kearney and CAF.

UNIJES Social Tech Challenge Winner — Bilbao, Spain | 2024
Voice-controlled wheelchair project

Academic Excellence Scholarship — Madrid, Spain | 2021, 2023, 2024
Top-scoring students across Community of Madrid

Baccalaureate Honorific Mention — Madrid, Spain | 2021
2nd highest GPA in Community of Madrid